EQUIPMENT PROPOSAL

4 TPH COOKED TUB PLANT — VAN BUREN, ARKANSAS

PROPOSAL SUMMARY

PREPARED FOR

Farmers Cooperative
Attn: Matt Crabtree, CEO
2105 Industrial Park Rd
Van Buren, Arkansas 72956

PROJECT

Process equipment for a 4 TPH cooked low-moisture tub plant
Equipment, engineering and start-up
230/460V/60Hz electrical
Ship to: Van Buren, AR

REFERENCE

MCE-Q-2610-FCO-R1
Basis: Matt Crabtree (Farmers Cooperative) discussion, Sep 2026; RangeLand 17 +Mag, 30-13 and 38 HI-E labels
Validity: 30 days from date of issue · Currency: USD
FOB MCE Stuart, FL or Newkirk, OK shop (per shop load)
Budgetary; firm price issued at end of FEED

Midwest Custom Engineering is pleased to propose a complete, standalone plant for cooked low-moisture protein tubs in the corner of your Van Buren warehouse: five tote unloaders and a loss-in-weight micro skid, a heated molasses tank, an MCE-designed 3-stage cook system, a rotary pour/cure turntable, a forced-air cure and cooling tunnel, a robotic stacker and stretch wrapper, and a plant control system with recipe batching and production reporting. MCE also provides detailed engineering and on-site start-up services. Installation is by others.

1. DESIGN BASIS

MCE engineering, Oct 20, 2026

PARAMETER	DESIGN VALUE	NOTES
Process	**Molasses + dry ingredients → 3-stage cook → rotary pour/cure → cure tunnel → robotic stack → wrap**	Low-moisture cooked molasses protein tubs; chemical-pour mineral tubs as Option B
Plant capacity	**4 TPH (8,000 lb/hr) rated**	Peak 5,000 x 225 lb tubs/month on one shift, about one tub every 1.8 min
Annual volume	**About 5,000 tons/yr**	About 44,400 tubs/yr at 225 lb equivalent
Products	**30-13, 38 HI-E and 17 +Mag type tubs**	Formulas set by an independent ruminant nutritionist during FEED; up to 20 stored recipes
Containers	**125, 225 and 250 lb tubs**	Plastic and fiber tubs; turntable nests and gripper sized for all three
Dry ingredients	**Budgetary: 5 tote stations + micro skid**	Station and bin count and weighing method set in FEED once formulas are final
Cook system	**MCE 3-stage jacketed cook**	Jacketed augers / mixing conveyors, 304 SS product contact
Heat source	**Packaged gas boiler or thermal-fluid heater**	Type and rating selected in FEED
Cure	**Turntable initial set + 100 ft forced-air tunnel**	Tunnel length confirmed in FEED from cure time
Location	**Corner of existing warehouse**	2105 Industrial Park Rd, Van Buren, AR (building approx. 49 m x 80 m)
Electrical	**230/460V / 3-phase / 60Hz**	Site voltage to be confirmed

2A. DRY INGREDIENT HANDLING – BUDGETARY ALLOWANCE

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
1	**Bulk-Bag (Tote) Unloader – Major Ingredients** • Budgetary allowance for tote stations handling major dry ingredients such as urea, magnesium oxide, phosphates, calcium carbonate and protein meal • Frame with bag-loading access, spout access chamber, flow promotion and metering screw discharge • Number of stations and ingredient assignment confirmed in FEED once final formulas are set	5	$19,745.00	$98,725.00
2	**Weigh System – Tote Stations** • Load cell weighing on each tote station, integrated with the plant control system • Weighing method (loss-in-weight or gain-in-weight) selected in FEED to suit the final formulas and cook control	5	$7,975.00	$39,875.00
3	**Electric Hoist & Trolley – Tote Loading** • Electric hoist and trolley at each tote station so one forklift operator can load totes	5	$5,975.00	$29,875.00
4	**Micro Ingredient Dosing** • Budgetary allowance for micro ingredient bins and feeders: trace mineral / vitamin premix and specialty additives • Number of bins, bin sizes and weighing method confirmed in FEED from final formulas	2	$23,940.00	$47,880.00
5	**Micro Skid Frame & Refill Hoppers** • Skid frame with refill hoppers and dust hoods for the micro bins; final configuration per FEED	1	$22,925.00	$22,925.00
6	**Collecting Screw Conveyor – Dry Ingredients to Cook Stage 1** • Collecting screw under the tote and micro feeders, discharging to cook stage 1 • 9" x 10 GA flanged trough, 14 GA cover, flush-end discharge; length per final layout • TEFC gearmotor with variable-frequency drive, 230/460V/3-phase	1	$20,310.00	$20,310.00
7	**Dust Collection – Tote Spouts & Micro Refill** • Bin vent filters and pickup hoods at the tote spouts and micro refill points, with exhaust fan • Collected dust returned to process	1	$15,960.00	$15,960.00

2B. MOLASSES RECEIVING & STORAGE

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
8	**Heated Molasses Storage Tank** • 10,000–12,000 gal vertical carbon steel tank, about four days of peak use; insulated and clad • Internal heating coil on the plant heat source, temperature transmitter, radar level transmitter, high-level alarm • Truck fill connection, manway, vent and overflow; final size confirmed in FEED	1	$69,815.00	$69,815.00
9	**Molasses Truck Unload Pump** • Gear-type truck unload pump with strainer and hose connection, TEFC motor, local start/stop	1	$13,375.00	$13,375.00

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
10	**Molasses Transfer Pump – Cook Feed** • Positive-displacement transfer pump with variable-frequency drive, pressure relief and gauges • Doses molasses to cook stage 1 by recipe rate	1	$14,535.00	$14,535.00
11	**Molasses Mass Flow Meter** • Coriolis mass flow meter with transmitter, tied to the plant control system for recipe dosing and batch totals	1	$15,960.00	$15,960.00
12	**Molasses Piping, Valves & Heat Tracing** • Insulated, heat-traced piping from tank to pumps and cook system • Isolation valves, strainer, cleanout and drain points; supports and hangers	1	$35,515.00	$35,515.00

2C. 3-STAGE COOK SYSTEM

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
13	**Jacketed Cook Stage – MCE Design** • Jacketed auger / mixing conveyor, 304 stainless steel product contact, heated jacket on trough • Stage 1 blends molasses and dry ingredients; stages 2 and 3 cook down to low moisture and condition the product • Bolted cleanout covers, vapor outlets, temperature transmitter at each stage discharge • General arrangement drawings issued for approval prior to fabrication	3	$64,415.00	$193,245.00
14	**Cook Stage Drive Package** • Shaft-mounted gearmotor with variable-frequency drive, one per stage, 230/460V/3-phase TEFC • Independent speed control sets residence time in each stage	3	$8,975.00	$26,925.00
15	**Heat Source – Packaged Boiler or Thermal-Fluid Heater** • Natural-gas-fired packaged steam boiler or thermal-fluid heater sized to the cook duty; type selected in FEED • Burner management system, circulation pumps or feedwater system, expansion tank, safety relief • Also serves the molasses tank heating coil and pipe tracing	1	$129,645.00	$129,645.00
16	**Heating Distribution – Cook Jackets** • Insulated steam or hot-oil supply and return piping to each jacket and to the molasses tank • Temperature control valves, isolation valves, traps or return lines	1	$31,510.00	$31,510.00
17	**Vapor Vent – Cook Stages** • Stainless hood and ductwork over the cook stage vapor outlets with exhaust fan • Carries moisture driven off during the cook out through the building sidewall	1	$15,725.00	$15,725.00
18	**Process Platform** • Structural steel platform over the cook stages and tote spouts, grating deck • Stairs, handrail, kick plate; shop primed and painted; final size from the approved layout	1	$60,995.00	$60,995.00

2D. ROTARY POUR / CURE TURNTABLE

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
19	**Rotary Pour / Cure Turntable** • MCE-built indexing turntable with tub nests for 125, 225 and 250 lb tubs • Empty-tub load station, pour station and transfer-out station; product takes its initial set while the table turns • Carbon steel structure, shop primed and painted; tub nest count set in FEED from set time	1	$61,630.00	$61,630.00
20	**Slew Bearing & Indexing Drive** • Heavy-duty slewing ring bearing with geared indexing drive, gearmotor and VFD • Position sensing at each station, interlocked with the pour valve	1	$29,885.00	$29,885.00
21	**Pour Valve & Check-Weigh Scale** • Positive-shutoff pour valve fed from cook stage 3, heated body • Scale platform under the pour station fills each tub to the recipe target weight; weights logged per tub	1	$29,885.00	$29,885.00

2E. CURE & COOLING TUNNEL

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
22	**Tunnel Conveyor Section, 10 ft** • Powered conveyor section carrying filled tubs from the turntable to the robotic stacker • Ten sections = 100 ft nominal; tunnel length set in FEED from the cure time • TEFC gearmotor drives, accumulation zone at the robot pick point	10	$8,875.00	$88,750.00
23	**Tunnel Enclosure Section, 10 ft** • Panelized enclosure over each conveyor section with hinged access doors • Supply plenum on one side, exhaust collection on the other	10	$3,945.00	$39,450.00
24	**Tunnel Supply Air Fan** • Supply fan blowing air through the tunnel to finish the cure and cool the tubs, with inlet filter and damper • TEFC motor, 230/460V/3-phase	2	$9,980.00	$19,960.00
25	**Off-Gas Exhaust Fan & Stack** • Exhaust fan pulling tunnel air and cure off-gas through ductwork to a sidewall stack • Outlet damper, weather hood; keeps off-gas out of the warehouse	1	$19,965.00	$19,965.00

2F. ROBOTIC STACKING & STRETCH WRAP

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
26	**Robotic Palletizer** • Remanufactured industrial palletizing robot, payload rated for a full 250 lb tub plus gripper • Robot controller and teach pendant; stacking patterns stored for each tub size • Pallet station with pallet locator and full-pallet discharge	1	$119,665.00	$119,665.00

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
27	**End-of-Arm Tub Gripper** • MCE-built gripper sized for 125, 225 and 250 lb tubs, pneumatic clamp with part-present sensing	1	$22,345.00	$22,345.00
28	**Robot Cell Safety** • Welded-wire perimeter guarding with interlocked access gate • Area scanners and pallet-station light curtain; safety relay integrated with the plant control system	1	$35,920.00	$35,920.00
29	**Stretch Wrapper** • Semi-automatic turntable stretch wrapper for finished pallets, powered pre-stretch film carriage	1	$29,885.00	$29,885.00

2G. PLANT CONTROLS

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
30	**Plant Control System / MCC** • PLC, 15" color touchscreen HMI at the cook system, operator pendants at the pour station and robot cell • Motor control center with starters, variable-frequency drives and disconnects for all MCE motors • Field instruments: temperature, level, weight and position; 24VDC control voltage	1	$129,645.00	$129,645.00
31	**Control Panel Build & Shop Test** • Panel assembly and wiring in the MCE shop, I/O checkout and simulated run before shipment	1	$32,380.00	$32,380.00
32	**PLC / HMI Programming & Integration** • Recipe batching for up to 20 formulas; loss-in-weight feeder, cook, turntable, tunnel and robot integration • Tub counter and per-tub weight log; production and ingredient-use reports by shift, day and month • Secure remote access for MCE support	1	$98,720.00	$98,720.00

2H. ENGINEERING & PROJECT MANAGEMENT

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
33	**Detailed Engineering & Project Management** • General arrangement in the warehouse corner, process flow and instrument diagrams, fabrication drawings • Electrical schematics and panel drawings; procurement, schedule and project management through start-up • Approval drawings issued to Farmers Cooperative before fabrication	1	$104,170.00	$104,170.00

2I. START-UP SERVICES

USD · sales tax excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
34	**Start-Up Services – Onsite Commissioning Assistance** • MCE start-up assistance and operator training on site • Equipment checkout, first product runs and recipe tuning with the nutritionist • Additional MCE days available at MCE's standard daily rate plus expenses	1	$13,800.00	$13,800.00

Dry Ingredient Handling – Budgetary Allowance – items 1–7	$275,550.00
Molasses Receiving & Storage – items 8–12	$149,200.00
3-Stage Cook System – items 13–18	$458,045.00
Rotary Pour / Cure Turntable – items 19–21	$121,400.00
Cure & Cooling Tunnel – items 22–25	$168,125.00
Robotic Stacking & Stretch Wrap – items 26–29	$207,815.00
Plant Controls – items 30–32	$260,745.00
Engineering & Project Management – item 33	$104,170.00
Start-Up Services – item 34	$13,800.00
Total — priced lines as quoted	**$1,758,850.00**

Payback at 5,000 tons/yr: about 11 months base case, 23 months conservative, on the MCE price. See Appendix A.

OPTIONS AND ADDERS

USD · sales tax excluded

OPT	DESCRIPTION	QTY	UNIT PRICE	NET ADDER
A	**Heated Fat Tank & Dosing** • Insulated, heated fat tank with transfer pump and mass flow meter for higher-fat formulas (38 HI-E type at 12% fat)	1	$38,640.00	$38,640.00
B	**Chemical-Pour Mineral Tub Package** • Setting-agent storage and metering pump, recipe logic for chemical-pour mineral tubs on the same turntable and tunnel	1	$36,950.00	$36,950.00
C	**Manual Stacking in lieu of Robotic Palletizer (items 26–28)** • Lift-assist tub stacking station with pallet locator at the tunnel exit; deletes robot, gripper and cell safety	1	$49,790.00	−$128,140.00
D	**Additional Tunnel Length, per 10 ft** • One added conveyor section and enclosure section (items 22–23) if FEED cure time requires a longer tunnel	1	$11,045.00	$11,045.00
E	**Spare Parts Package** • Recommended first-year spares: screw flighting and bearings, pour valve seals, load cell, VFD, robot gripper pads	1	$18,760.00	$18,760.00

3. FURNISHED BY OTHERS

- Building modifications, roof and wall penetrations other than MCE vent and stack thimbles, ventilation and fire protection
- Concrete, foundations and anchor bolts; tank pad
- Utilities to the plant battery limits: natural gas service, electrical service and transformer, water and compressed air
- Air, building and boiler permits (MCE provides equipment data for applications)
- Nutritionist services, final formulations and feed registrations
- Raw materials, tubs, lids, labels, pallets and stretch film; forklifts
- Installation, millwright, rigging and crane, electrical installation and labor
- Freight from the MCE shop (quoted at time of shipment)
- Sales and use taxes

4. FREIGHT, DELIVERY AND SCHEDULE

ITEM	BASIS
FEED	Fixed-price front-end engineering, 6–8 weeks, $50,000–$75,000, credited in full against the plant price at order. Produces final formulas with the nutritionist, layout, tunnel length, heat source selection and firm price.
Freight	FOB origin, MCE Stuart, FL or Newkirk, OK shop (assigned at order based on shop load and demand); freight to Van Buren, AR quoted at time of shipment, or shipped freight collect.
Delivery	About 9–12 months from FEED start to commissioning. Long-lead items (heat source, robot, feeders) ordered at release. Approval early 2027 targets start-up before the fall 2027 tub season.
Approval drawings	General arrangement and plant layout issued for Farmers Cooperative approval before fabrication.
Installation	By others. MCE start-up services and operator training included (item 34); additional MCE days at MCE's standard daily rate plus expenses.

5. COMMERCIAL TERMS

Payment: 50% down with order, balance before shipment. Payment terms are as stated on this quote and on the invoice. Past-due balances may be reported to business credit bureaus, including Dun & Bradstreet and Experian. Prices are in US dollars and exclude sales tax, freight, duties and installation. **Sales tax:** this project is in Arkansas – Arkansas sales tax will be added at invoicing unless a valid Arkansas exemption certificate is provided before invoicing. Pricing is valid 30 days from the date of this proposal. Warranty per MCE standard terms and conditions.

Uptime support: MCE's goal is maximum uptime for every plant we build. A separate maintenance and parts agreement covering the plant's core components – cook stages, turntable, tunnel conveyors, feeders and drives – is available on request, with scheduled inspections, priority parts and remote support through the plant control system.

PREPARED BY

Jason Shipley

President, Midwest Custom Engineering, Inc.

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FOR FARMERS COOPERATIVE

Accepted: _____

Name / Title: _____

Date: _____ PO #: _____

APPENDIX A. ESTIMATED FORMULATIONS AND PAYBACK

Estimates for budgeting

Batch formulas built to the guaranteed analysis on comparable RangeLand labels. Ingredient prices are delivered estimates for October 2026. A ruminant nutritionist sets the final formulas during FEED.

INGREDIENT	$/TON	30-13 %	38 HI-E %	17 +MAG %
Cane molasses	$240	52.0	44.5	52.0
Soybean meal 48%	$340	10.0	9.0	12.0
Cottonseed meal 41%	$320	8.0	7.0	6.0
Feather meal	$550	6.0	6.0	4.0
Distillers grains (DDGS)	$190	3.0	3.0	6.0
Urea, feed grade	$550	4.6	7.1	0.0
Fat, animal or vegetable	$1,000	4.0	11.0	4.0
Dicalcium phosphate	$850	5.5	5.5	5.5
Monocalcium phosphate	$900	1.5	1.5	1.5
Calcium carbonate	$60	1.5	1.5	0.5
Magnesium oxide	$750	1.9	1.9	6.5
Sodium hydroxide 50%	$700	0.5	0.5	0.5
Trace mineral / vitamin premix	$2,000	1.5	1.5	1.5
Batch cost, $/ton		**$397.30**	**$456.45**	**$401.00**
Finished cost after cook (90% yield), $/ton		**$441.44**	**$507.17**	**$445.56**
Ingredients per 225 lb tub		**$49.66**	**$57.06**	**$50.12**

PER 225 LB TUB, BASE CASE	30-13	38 HI-E	17 +MAG
Share of volume	50%	25%	25%
Ingredients	$49.66	$57.06	$50.12
Tub, lid and label	$14.00	$14.00	$14.00
Labor (4 operators, one shift), energy and maintenance	$7.43	$7.43	$7.43
Cost to make	**$71.09**	**$78.49**	**$71.55**
Current buy price (assumed – replace with Farmers Cooperative cost)	$115.00	$125.00	$105.00
Savings per tub	**$43.91**	**$46.51**	**$33.45**

PAYBACK ON THE MCE PRICE	BASE CASE	CONSERVATIVE
MCE price	$1,758,850	$1,758,850
Savings per tub, mix-weighted	$41.94	$20.58
Annual savings (44,444 tubs)	$1,864,205	$914,754
Simple payback	**11.3 months**	**23.1 months**
5-year net savings, after the MCE price	$7.6M	$2.8M

Conservative case: ingredient prices +10%, buy prices 13% lower ($100 for a 30-13 tub), $15 tubs, $1.25/tub energy. Retail for a 225 lb RangeLand 30-13 tub ran $125–$200 in October 2026 (NRS, Tractor Supply). Excludes installation by others, building, utilities to the plant, working capital and taxes. Interactive calculator: claude.ai/artifact/1LicacHXU4geYwCxifujn7